Worksheet 6.2

Science Quest 9: pages 240–1

How big is an atom?

Answers

It is hard to visualise the size of tiny objects, like atoms and their nuclei. However, making comparisons with something we are more familiar with sometimes helps.

Use an everyday object to create an image that will help people to understand how small the nucleus of an atom is compared with the size of the entire atom. Choose from the suggestions provided or use your own ideas. Atoms vary in size, but for the purpose of this activity, assume the nucleus of an atom is 100 000 times smaller than the size of the entire atom.

(a) If a marshmallow represents the nucleus of an atom, how big would the entire atom be? (*Hint:* You will need to measure the diameter or the length of a marshmallow.)

Marshmallows vary in size. In this calculation, the marshmallow is 3 cm long. If the nucleus is 3 cm on this scale, then the entire atom is 100 000 times larger.

= 3 100 000
= 300 000 cm
= 3000 m
= 3 km

(b) If one gradation on millimetre-lined A4 graph paper represents the nucleus of an atom, how many sheets of graph paper would you need to glue together to represent the entire atom?

If the nucleus is 1 mm, then the diameter of the whole atom would be:

= 1 100 000
= 100 000 mm
= 100 m.

A sheet of A4 graph paper is 25 cm long, which is 0.25 m.

Since 100 ÷ 0.25 = 400, 400 sheets of A4 graph paper would need to be glued together to represent the size of the atom.

(c) If a marble, representing a nucleus, is placed in the centre of a sports ground, how big would the sports ground need to be to represent the entire atom?

If the marble nucleus has a diameter of 15 mm, then the diameter of the sports ground is 100 000 times the marble's diameter.

= 15 100 000
= 1 500 000 mm
= 1500 m
= 1.5 km

Therefore, the sports ground would need to have a diameter of 1.5 km.



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